

To-Do List Application

Project Overview

The **To-Do List Application** is a Python console-based task management system that helps users organize their daily activities. It allows users to add, view, update, delete, and mark tasks as completed through a simple menu-driven interface.

This project is ideal for beginners learning Python fundamentals such as lists, dictionaries, loops, conditional statements, exception handling, and user input.

Features

- Add new tasks
 - View all tasks
 - Update existing tasks
 - Delete tasks
 - Mark tasks as completed
 - Track task status (Pending / Completed)
 - Handles invalid input
 - Easy-to-use menu-driven interface
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Technologies Used

- Python 3.x
 - Python Standard Library
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Project Structure

```
To-Do-List-Application/  
├── todo_list.py  
├── README.md  
└── LICENSE
```

How to Run

Step 1

Install **Python 3.8** or later.

Step 2

Clone or download this repository.

Step 3

Open a terminal or command prompt.

Step 4

Navigate to the project folder.

```
cd To-Do-List-Application
```

Step 5

Run the application.

```
python todo_list.py
```

Menu

```
=====
                        TO-DO LIST APPLICATION
=====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Exit
=====
```

Sample Output

```
Enter your choice (1-6): 1
Enter new task: Complete Python Project
☒ Task added successfully!
```

Example Task List

```
----- YOUR TASKS -----
```

1. Complete Python Project [Pending]
 2. Upload Project to GitHub [Completed]
 3. Practice Python [Pending]
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Error Handling

The application includes:

- Prevents invalid menu choices.
 - Handles invalid task numbers.
 - Displays appropriate messages when no tasks are available.
 - Uses exception handling to prevent program crashes.
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Future Improvements

- Save tasks to a file (JSON/CSV)
 - Load tasks automatically on startup
 - Task due dates
 - Task priorities (High, Medium, Low)
 - Search tasks
 - Sort tasks
 - GUI version using Tkinter
 - Database integration (SQLite/MySQL)
 - Dark mode interface
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Python Concepts Used

- Lists
 - Dictionaries
 - Loops (`while`, `for`)
 - Conditional Statements (`if-elif-else`)
 - Exception Handling (`try-except`)
 - Functions
 - User Input
 - String Formatting
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Learning Objectives

This project helps learners understand:

- CRUD (Create, Read, Update, Delete) operations
 - Data storage using lists and dictionaries
 - Exception handling
 - Menu-driven programming
 - User interaction in console applications
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License

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